

## Test Case Planning

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Github: <https://github.com/Yuplox/CMSC-355-Project.git>

### 1. Use Case 3 - AI Diet & Workout Suggestions

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| Test Case ID     | TC01 - Generate Diet Plan                                                                                                                                                                                                                                                                                                       |
| Test Objective   | Verify that the AI generates a valid diet plan when complete user information is provided.                                                                                                                                                                                                                                      |
| Preconditions    | User is on the correct page for AI diet suggestions                                                                                                                                                                                                                                                                             |
| Test steps       | <ol style="list-style-type: none"><li>1. Open the AI Diet Suggestions page.</li><li>2. Enter the user age.</li><li>3. Enter the user height.</li><li>4. Enter the user weight.</li><li>5. Select the activity level.</li><li>6. Select the goal (lose weight, gain muscle, maintain).</li><li>7. Click Generate Plan.</li></ol> |
| Input Values     | Int age = 21<br>Int height = 170<br>Int weight = 75<br>String activity_level = "Moderate"<br>String goal = "Lose Weight"                                                                                                                                                                                                        |
| Expected results | A diet plan is generated and displayed with calorie targets and recommended meals. No error messages appear.                                                                                                                                                                                                                    |

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| Test Case ID | TC02 - Generate Workout Plan |
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| <b>Test Objective</b>   | <b>Verify that the system generates a workout plan based on the user's selected fitness goal.</b>                                                                                                                         |
| <b>Preconditions</b>    | User is on the correct page for AI workout suggestions.                                                                                                                                                                   |
| <b>Test steps</b>       | <ol style="list-style-type: none"><li>1. Open the AI Workout Suggestions page.</li><li>2. Select a workout goal.</li><li>3. Select the number of workout days per week.</li><li>4. Click Generate Workout Plan.</li></ol> |
| <b>Input Values</b>     | String goal = "Build Strength"<br>Int days_per_week = 4                                                                                                                                                                   |
| <b>Expected results</b> | A workout plan is displayed with exercises assigned across the selected number of days.<br>Exercises match the selected fitness goal.                                                                                     |

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| <b>Test Case ID</b>     | <b>TC03 - Invalid or Missing Input</b>                                                                                                                                                       |
| <b>Test Objective</b>   | <b>Verify that invalid or missing user input is rejected and an error message is displayed.</b>                                                                                              |
| <b>Preconditions</b>    | User is on the AI suggestions page.                                                                                                                                                          |
| <b>Test steps</b>       | <ol style="list-style-type: none"><li>1. Open the AI suggestions page.</li><li>2. Leave one or more required fields empty OR enter invalid values.</li><li>3. Click Generate Plan.</li></ol> |
| <b>Input Values</b>     | Int age = -5<br>Int height = 170<br>String weight = ""<br>String activity_level = "Moderate"<br>String goal = "Maintain"                                                                     |
| <b>Expected results</b> | Plan is not generated and the system displays a warning message describing which fields are invalid or missing.                                                                              |

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| <b>Test Case ID</b>   | <b>TC04 - Regenerate Plan After Changing Preferences</b>                                              |
| <b>Test Objective</b> | <b>Verify that updating user preferences generates a new plan that differs from the previous one.</b> |
| <b>Preconditions</b>  | A valid AI generated plan already exists.                                                             |
| <b>Test steps</b>     | <ol style="list-style-type: none"><li>1. Generate an initial plan.</li></ol>                          |

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|                         | <ol style="list-style-type: none"><li>2. Change at least one preference (goal or activity level).</li><li>3. Click Generate Plan again</li><li>4.</li></ol> |
| <b>Input Values</b>     | String original_goal = "Lose Weight"<br>String new_goal = "Gain Muscle"                                                                                     |
| <b>Expected results</b> | A new plan is shown reflecting the updated preferences. The new plan differs from the original and no duplicate entries are created.                        |